Assignment – 05:

Problem Statement:

Write a program in C to perform basic matrix operations, such as:

1. Addition of two matrices
2. Saddle point of a matrix
3. Inverse of a matrix
4. Magic square of a matrix

Code:

```
#include <stdio.h>
int main()
{
    int A[10][10],B[10][10],C[10][10];
    int i,j,r,c,n;
    printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c);
    printf("Enter Matrix A:\n");
    for (i=0;i<r;i++)
    {
        for (j=0;j<c;j++)
        {
            scanf("%d",&A[i][j]);
        }
    }
    printf("Enter Matrix B:\n");
    for (i=0;i<r;i++)
    {
        for (j=0;j<c;j++)
```

```
    {
        scanf("%d",&B[i][j]);
    }
}
printf("Addition:\n");
for (i=0;i<r;i++)
{
    for (j=0;j<c;j++)
    {
        C[i][j]=A[i][j]+B[i][j];
        printf("%d ",C[i][j]);
    }
    printf("\n");
}
int found=0;
for (i=0;i<r;i++)
{
    int min=A[i][0],col=0;
    for (j=1;j<c;j++)
    {
        if (A[i][j]<min)
        {
            min=A[i][j];
            col=j;
        }
    }
```

```
    int flag=1;
    for (int k=0;k<r;k++)
    {
        if (A[k][col]>min)
        {
            flag=0;
            break;
        }
    }
    if (flag)
    {
        printf("Saddle Point = %d\n",min);
        found=1;
    }
}
if (!found)
{
    printf("No Saddle Point\n");
}
int M[2][2];
float det;
printf("Enter 2x2 matrix:\n");
for (i=0;i<2;i++)
{
    for (j=0;j<2;j++)
    {
```

```
            scanf("%d",&M[i][j]);
        }
    }
    det=M[0][0]*M[1][1]-M[0][1]*M[1][0];
    if (det==0)
    {
        printf("Inverse not possible\n");
    }
    else
    {
        printf("Inverse:\n");
        printf("%.2f %.2f\n",M[1][1]/det,-M[0][1]/det);
        printf("%.2f %.2f\n",-M[1][0]/det,M[0][0]/det);
    }
    int sum=0,temp;
    printf("Enter order of square matrix: ");
    scanf("%d",&n);
    printf("Enter matrix:\n");
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            scanf("%d",&A[i][j]);
        }
    }
    for(j=0;j<n;j++)
```

```
{
   sum+=A[0][j];
}
for(i=0;i<n;i++)
{
   temp=0;
   for(j=0;j<n;j++)
   {
      temp+=A[i][j];
   }
   if(temp!=sum)
   {
      printf("Not Magic Square\n");
      return 0;
   }
}
for(j=0;j<n;j++)
{
   temp=0;
   for(i=0;i<n;i++)
   {
      temp+=A[i][j];
   }
   if(temp!=sum)
   {
      printf("Not Magic Square\n");
```

```
            return 0;
        }
    }
    temp=0;
    for(i=0;i<n;i++)
    {
        temp+=A[i][i];
    }
    if(temp!=sum)
    {
        printf("Not Magic Square\n");
        return 0;
    }
    temp=0;
    for(i=0;i<n;i++)
    {
        temp+=A[i][n-i-1];
    }
    if(temp!=sum)
    {
        printf("Not Magic Square\n");
        return 0;
    }
    printf("Magic Square\n");
    return 0;
}
```

Input & Output:

```
Enter rows and columns: 2 3
Enter Matrix A:
3 4 5 2 5 6
Enter Matrix B:
3 6 7 9 7 1
Addition:
6 10 12
11 12 7
Saddle Point = 3
Enter 2x2 matrix:
4 6 1 6
Inverse:
0.33 -0.33
-0.06 0.22
Enter order of square matrix: 3
Enter matrix:
8 1 6 3 5 7 4 9 2
Magic Square
```